Form 62 Mar. 27 Apr. 2027 Synthesis problems crossing several fields of the programme

1. Warming · Flash Solve 5 $x - 3 = 2x + 9$.
2. Heart · Problem A price rose by 20 % and then a fixed rebate of 1 000 Ar and was then worth 11,000 Ar. What was the initial price?
3. Heart + · Method $f(x) = 2x + 1$ and $g(x) = -x + 7$. Find the intersection point of the curves then resolve $f(x) < g(x)$.
4. Back · Flash A recipe for 4 people requires 250 g of flour. How much for 6 People?
5. Red thread · Flash A figure of 4 cm ^{2} is enlarged with a ratio 3. enlargement?

Form 63 game. 29 Apr. 2027 Reasoning by the absurd and by disjunction of cases

1. Warming · Flash What is the parity of $n(n+1)$ for an integer n ?
2. Heart · Disjunction of cases Show that $n(n+1)$ is even for the integer n.
3. Heart + · Absurd Show by the absurd that the sum of a rational r and an irrational a is Irrational.
4. Back · Formula $g(x) = 2x + 5$. Image of -1 ? Preimage of 11 ?
5. Red thread · Formula → graph Place the dots off $-2\,;\,-1\,;\,0\,;\,1\,;\,2$ of the curve of $f(x) = x^2$; give their orders.

Form 64 Mon. 3 May 2027 Logical, contrapositive and counter-example connectors in situation

1. Warming · Flash Write the denial of " $x > 2$" and then " $x > 2$ or $x < -1$ ».
2. Heart · Method Write the contraposition of "if n^2 is even, so n is even".
3. Heart + · Counterexample Refuse: "any prime number is odd."
4. Return · True/false True or false? Justify. The perimeter of a square is proportional to the length of His side.
5. Red thread · Code reading What's sent back $h(3)$? Write $h(x)$ in the form of an expression.

Probabilities: Packing and weighted trees

Form 65 Mar. 4 May 2027 Probabilistic model and equiprobability

1. Warming · Flash We're throwing a balanced piece three times.
2. Heart · Method Unfixed: $P(6) = 0.25$ and the other five sides are equiprobable. $P(1)$.
3. Heart + · Critical Spirit "It will rain or it will not rain tomorrow: so there is one in two chances." Is equiprobable model justified?
4. Back · Flash Which figures are composed of a pyramid pattern with a square base?
5. Red wire · Translate A side square x see each of its sides increase by 3. Express the increase in relation to the x.

Form 66 Mon. 10 May 2027 Law of large numbers (vulgarized version): simulation

1. Warming · Flash Out of 100 launches, 47 "cells". What is the frequency of "cells"?
2. Heart · Interpret Why, on 10 throws of a balanced piece, can the frequency of "cell" What's going on with 10,000 shots?
3. Heart + · True/false True or false? Justify. By launching 1,000 times a balanced piece, we get exactly 500 "cells".
4. Back · Problem Price incl. $P(x) = 1{,}2\,x$ (20% VAT). $P(250)$ and the preimage of 480.
5. Red thread · Method Solve in $\mathbb{R}$: $\dfrac{2x-1}{3} = \dfrac{x+4}{2}$.

Form 67 March 11 May 2027 Python simulation of a repeated random experiment

1. Warming · Flash $P(A) = 0.35$. Calculate $P(\overline{A})$.

2. Heart · Code reading What does this program calculate? What value will it display about?

```
from random import randint
account = 0
for i in range(1000):
        if randint(1, 6)==6:
                account = account + 1
print(account / 1000)
```

3. Heart + · Method Balanced di 12 faces numbered from 1 to 12. Likelihood of getting a number First?

4. Back · Construction programme M belongs to the mediator of $[AB]$. What can be said of MA and MB ? How can I build it on the compass?

5. Red thread · Recognize What is the nature of the section of a cube by a plane parallel to one of its Faces?

Form 68 13 May 2027 Conditional likelihood P $_A(B)$

1. Warming · Flash Of the 25 half boarders, 15 are girls. Like it's a girl?

2. Heart · Method In a class of 30 students, 12 play football, 5 of which also play hand. selects a student at random: calculate $P_F(H)$, F : "Play football", H : "Play the Hand".

3. Heart + · Method In an equiprobable model, $\text{Card}(A) = 50$ and $\text{Card}(A \cap B) = 20$. Calculate $P_A(B)$.

4. Back · Flash Store in ascending order: $\frac{7}{4}, -\frac{3}{2}, 0{,}5$.

5. Red thread · True/false True or false? Justify. A quadrilateral with two opposite sides parallel is a parallelogram.

Form 69 March 18 May 2027 Conditional likelihood and cross-table

1. Warming · Flash In the following table, what is the total of the "sport" line?

	Girls	Boys	Total
Sport	18	22	…
No sports	12	8	20
Total	30	30	60

2. Heart · Method We pick a student at random from this table. $P(S)$, $P_F(S)$ and $P_S(F)$ (F : girl, S : sport).

3. Heart + · True/false True or false? Justify. $P_F(S) = P_S(F)$.

4. Back · Flash Calculate $5^0 + 2^{-1}$.

5. Red thread · Method $ABCD$ is a parallelogram with $A(1;1)$, $B(5;2)$, $C(6;5)$. Determine D.

Form 70 game. 20 May 2027 Weighted trees: construction

1. Warming · Flash What is the value of the sum of the probabilities of branches from the same node?

2. Heart · Method In a population, 30% have a disease M. A test is positive (T) for 90% of patients and 5% of non-malads.
I'm a tree.

3. Heart + · Translate Translate into symbols: " 90% of patients have a positive test".

4. Back · Flash Calculate $\frac{2}{3} \div \frac{4}{9}$.

5. Red thread · Problem Three headlights flash every 6, 8 and 9 seconds. $t = 0$. When the next simultaneous flashing?